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Cognitive Decline as Experienced in *The Father* (2020) and *Vortex* (2021)

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Introduction

How do films make the complex and incomprehensible experience of cognitive loss perceptible to viewers? Dementia, marked by memory loss, confusion, and the gradual fading of one's sense of self, poses a unique challenge for films. This paper will examine how the films *The Father* (directed by Florian Zeller, 2020) and *Vortex* (directed Gaspar Noe, 2021) approach this challenge, exploring how they use filmic strategies to immerse the viewer in the cognitive and emotional disarray of cognitive loss. Rather than simply telling stories about memory loss, both films draw the audience into a fragmented, disoriented experience that mirrors the internal world of their characters who struggle with dementia, creating a sense of uneasy closeness with minds in decline.

Drawing on both film theory and cognitive approaches to spectatorship, this paper situates *The Father* and *Vortex* within a cognitive- aesthetic framework that emphasizes emotional immersion, spatial disorientation, and narrative rupture. *The Father* disorients the viewer through shifting environments and casting changes that simulate memory failure and the loss of temporal continuity. In contrast, *Vortex* employs a split- screen and (mostly) real time duration to generate perceptual and emotional fatigue, mimicking the isolating simultaneity of two deteriorating subjectivities. While the formal and aesthetic choices in each film differ dramatically, both seek to portray the internal logic of dementia, translating the intangible syndrome into visible cinematic forms. Together, these films highlight how cinema can function not only as a window into cognitive impairment but as a tool for simulating its felt experience.

Theoretical Framework of Cognitive Decline in Film

David Bordwell's concept of "part-time cognitivism" provides a useful entry point for analyzing viewer engagement in both films. According to Bordwell, spectators approach films through a combination of rational interpretation and emotional immersion, working through disjunctions and ambiguities with both mental effort and affective investment. (Bordwell 6) Noel Carroll similarly emphasizes the importance of emotional engagement in narrative comprehension:

Our experience of actions and events in movies differs radically from our normal experiences; movie actions and events are so organized, so automatically intelligible, and so clear. The arresting thing about movies, contra realist theories, is not that they create the illusion of reality, but that they reorganize and construct, through variable framing, actions and events with an economy, legibility, and coherence that are not only automatically available, but which surpass, in terms of their immediately perceptible basic structure, naturally encountered actions and events. Movie actions evince visible order and identity to a degree not found in everyday experience. This quality of uncluttered clarity gratifies the mind's quest for order, thereby intensifying our engagement with the screen. (Carroll 93)

From the perspective of cognitive neuroscience, Jeffrey Zacks and Joseph Magliano describe how narrative comprehension involves memory, prediction, and segmentation- capacities that are profoundly disrupted in portrayals of dementia. (Zacks & Magliano 4).

There is good behavioural and neurophysiological evidence that when people watch ongoing activity, they segment it into meaningful events. Segmentation is an ongoing concomitant of normal perception—one that is related to eye movements and blinks, but that reflects the operation of a broader system for orienting attention and updating memory. (Zacks & Magliano 4)

Ed Tan's work on the "Psychology of the Film" furthers this approach by suggesting that emotional responses are triggered when the viewer encounters narrative dissonance, creating a unique kind of viewer engagement through cognitive tension.

Once imagination takes over from perception, the reality status of stimuli is traded for believability. As part of the imagination fictional events are matched with higher order genre specific narrative schemas, and then dealt with as possibilities in a particular world.
(Tan 11)

Phenomenological film theorists such as Julian Hanich and Christian Ferencz- Flatz shift the focus from narrative processing to bodily perception. For them, the cinematic experience includes a visceral, affective dimension, one that is especially relevant in films that aim to simulate mental deterioration (Hanich and Ferencz- Flatz 12). Similarly, Ralley and Connolly highlight the notion of "presence" in film, suggesting that viewers can be drawn into cognitive- emotional states that mirror on- screen experiences through visual manipulation. "Emotional and intentional attunement to others means that a bridge to others, a condition for presence, is readily available."
(Ralley and Connolly 53)

Cognitive Science of Dementia

Dementia in film is often rendered through metaphorical and perceptual distortions. Scholars have noted how weather metaphors (fog, clouds, rain) are used to describe mental decline, symbolizing confusion and loss of clarity (Schweda 4). But underneath cinematic metaphors, is a syndrome of cognitive decline.

Dementia is defined by chronic, acquired loss of two or more cognitive abilities caused by brain disease or injury. [...] Memory requires the recording, storage, and retrieval of information. The most common clinical presentation of [Dementia] is a slow onset and gradually progressive loss of memory, typically with inability to learn new information and particularly autobiographical information, such as recent events in one's life. This is because [Dementia] preferentially affects brain networks involved in episodic memory. (Arvanitakis 3)

This gradual erosion of episodic memory leads not only to forgetting past experiences but also to the disruption of one's sense of identity. As Stan Klein and Shaun Nichols emphasize, memory is essential to constructing and maintaining the self; its breakdown results in a fragmented and unstable personal narrative. (Klein and Nichols 677) Beyond memory, dementia also disrupts executive functioning, working memory, and spatial- temporal orientation. These impairments make everyday decisions, attention, and basic navigation difficult, contributing to a loss of autonomy and a destabilized relationship with one's surroundings. (Klein and Nichols 685)

Medical literature emphasizes how dementia disrupts continuity of identity, spatial awareness, and the ability to place oneself in time, leading to a fragmented, often terrifying subjective world.

The onset of dementia involves a great variety of feelings. Many experience feelings of fear, insecurity, confusion or disbelief when they start to realize "something is wrong". This might even cause them to feel like they are "going mad." While for some having

dementia brings confusion, shock, sorrow or anxiety, others experience strong feelings of discontent like annoyance, anger, distress or frustration (de Boer 1028- 1029)

These insights align with the narrative and visual strategies deployed in *The Father* and *Vortex*, which prioritize disorientation as a central stylistic device. Beyond memory loss, dementia often involves anosognosia (a lack of awareness of one's cognitive deficits) and agnosia (the inability to recognize familiar objects, faces, or environments). (Arvanitakis 4- 5) These symptoms contribute to a perceptual alienation from the world, turning the familiar into something eerie and unplaceable. This is paralleled in *The Father* and *Vortex*, where familiar domestic spaces are rendered unfamiliar, confusing, and even threatening. In *The Father*, slight changes to a flat's layout echo the neural fragmentation that impairs place recognition. In *Vortex*, the split screen mimics divided attention and confusion.

The Father (2020): A Cognitive Maze

Florian Zeller's *The Father* (2020) presents dementia not as a syndrome to be observed, but as an experience to be inhabited. The narrative unfolds through the perspective of Anthony, an elderly man struggling with dementia living in London. Anthony's sense of reality is destabilized by recurring scenes, contradictory character identities, and subtly altered environments. For example, throughout the film, characters change appearance and are played by an entirely different actor. This happens when Anthony finds out that he is no longer living in his flat, but at his daughter's, Anne's, home. When she returns home, he wants to confront her but as she turns around- he discovers she's a different person- played by an entirely different actress. In another instance, Anthony meets a nurse who he gets along with, Laura, after struggling to get along and feel comfortable with several nurses. They get along and have great conversations so when Laura is about to arrive for the day, he is excited. However, when she arrives, she is played by a different actress. All this impacts Anthony, as he begins to feel like he is "going mad" and begins to have an emotional breakdown.

The use of recurring locations (especially the flat where most of the film takes place) serves as a spatial puzzle. Small variations in furniture, wall colors, and lighting destabilize the viewer's spatial cognition, echoing Anthony's own struggle to orient himself. The location of the filming was mostly in one singular apartment. However, small details change throughout the film in the apartment to signify that this is not the same flat. At first the location is Anthony's flat, then Anne, then it's Anthony's flat again, and then it is a nursing home room. The changes are subtle, in the beginning of the film his walls are painted light yellow and he has a wooden piano against the wall. In Anne's flat, it's the same location and room but the walls are painted pastel blue and against the wall in a whisky bar. Once Anthony returns to his flat (something he is unaware he left in the first place) the decor is still the decor of his daughter's apartment but he knows it is his- causing further confusion. This spatial disorientation parallels real- life difficulties experienced by dementia patients, who often mistake one location for another or believe they are in a former

residence. Zeller's decision to maintain a single location with morphing details mirrors the breakdown of the cognitive memory.

This approach is supported by Daniel Levin and Caryn Wang's work on spatial representation in film, which shows how disrupted spatial logic can produce cognitive discomfort and confusion but also go past unnoticed- both true in Anthony's case. The film weaponizes these disruptions to mirror neurological fragmentation. For the viewer, as for Anthony, changes may register subconsciously, generating unease without immediate clarity. This cinematic strategy draws on the cognitive phenomenon of 'change blindness,' a failure to detect alterations in visual stimuli when not directly attended to, demonstrating the fallibility of both perceptual processing and memory.

This kind of disruption can bring to the fore two fundamental issues of visual cognition. First, in forcing the audience to focus on one idiosyncratic piece of the visual world, these insert shots make clear the degree to which our awareness of specific visual objects and properties is highly selective. This kind of selectivity has been the focus of much research documenting the surprising degree to which we are aware of far fewer visual properties than one might think. For example, a phenomenon referred to as "change blindness" occurs when people fail to detect visual changes that occur across views. (Levin and Wang 24- 25)

In *The Father*, change blindness is not only a tool for disorientation but also a cognitive metaphor: it reflects the tragic disconnection between what Anthony perceives and what the world actually presents. The subtle reconfigurations of space and people reinforce the idea that cognition is always vulnerable to distortion.

The film employs subjective camera work to put the audience into the shoes of Anthony's lack of understanding of his surroundings and reality. Towards the end of the film, Anthony is overwhelmed and confused and walks into his room in his flat. The camera locks into a close up shot of Anthony as he sits on his bed and tries to make sense of everything. Then, we hear his

daughter ask him if he likes the room. We are still on the same shot of his close up- still seemingly in his bedroom, in his flat. But, with a cut to a long shot, to a nursing home room and a nurse standing behind them, it is clear that the audience along with Anthony experienced gaps in events.

This subjective cinematography invokes what Grodal describes as empathetic engagement, where the viewer's affective and perceptual systems are recruited to match the character's internal state.

In order to assimilate a character's situation, the viewer must have a sense of the character's "position". Sensing the latter is an embodied process and derives from watching the character's actions, her point of view as structured and framed by the film, as well as imagining or directly experiencing. Engagement is fed by the viewer's own previous experiences of pain and loss, which can influence the 'like me' framework – depending on the relationship between viewer and character. (Grodal 26-27)

The film's withholding of spatial context and continuity editing enhances this alignment, immersing us directly into Anthony's experience of confusion, misplacement, and vulnerability.

***Vortex* (2021): Duality of Decay**

On the other hand, Gaspar Noé's *Vortex* (2021) opts for raw, observational realism. The film is structured around a split screen that simultaneously follows an elderly couple (Elle and Lui), each descending into their own cognitive and physical decline. This division of the screen functions as a metaphor for a fragmented consciousness.

The split- screen reinforces the idea of a mind that is deteriorating, getting lost in the labyrinthine décors of the film and where, through the fragmentation of space on screen, it may also be formally lost. (Simoes 70)

The viewer is deliberately cognitively overloaded, forced to split attention between two scenes at once, simulating the manic simultaneity and incoherence often experienced in dementia.

“Representation of dementia via the split screen brings relation and interdependence into relief, revealing the reality of a broader vulnerability and relationality shared by everyone.” (Taliyan 619)

The split screen appears a few minutes into the film. Elle and Lui sleep in bed. Suddenly, Elle wakes up and stares at the ceiling in silence. As she does, a vertical line slowly descends, dividing the screen between Elle and Lui and initiating the split screen perspective. The sequence appears in one uncut shot- much like much the rest of the film. The film's long takes and minimal editing create a sense of temporal realism.

[The synchronization of the shots is] fundamental for managing two concurrent entities in time, resulting in the coordinated mise en scène of the multi- protagonists on the screens. (Simoes 69)

The viewer is locked into real- time observation and is denied the usual narrative skips that allow for mental rest or narrative closure. This slow, gruelling pace mirrors the agonizing slowness of cognitive deterioration. The film's visual aesthetics are unpolished with low lighting, natural

sounds, and handheld camera. Hanich and Ferencz- Flatz describes this kind of aesthetic as “phenomenological realism.”

Phenomenology is an attempt to film, in *slow motion*, that which has been, owing to the manner in which it is seen in natural speed, not absolutely unseen, but missed, subject to oversight. (Hanich and Ferencz- Flatz 19)

In another scene, Elle wanders off to go to the store but forgets where she is or what she came there to do. She is left confused at a local shop and freezes in place. Meanwhile, Lui realizes she has left the apartment and begins to look for her. This scene unfolds in two simultaneous, continuous shots that do not cut until Lui finds Elle. When they reunite, both sides of the screen show the same image, signifying restored unity and physical safety. The handheld camera follows both characters as they move from location to location.

Hand- held camera and steadicam shots elicit stronger mirroring responses and an augmented sense of involvement in the scene because these types of cinematography more closely resemble human movement than static camera, zooms, or dollymounted tracking shots. The findings of this neuroscientific study suggest camera movement, teamed with the movement of the figures on screen, can function to elicit a sense of affective identification. This shared perceptual experience may elevate empathic concordance with observed actions because audience members feel as though they are located with the character. (Stadler 458)

Thus, the viewer is drawn not only mentally but also physically into the scene, experiencing dementia as a cognitive state rather than a narrative one.

Vortex offers no stable point of reference, no authoritative narrative perspective. Instead, it presents a world in which subjectivity is collapsing, and the viewer must constantly recalibrate their understanding of events and motivations.

There is no fixed relationship between a given character and a given half of the screen. Whom the image is associated with shifts. Relations are a widely cast web and, in the dynamic between the two images, the film reveals us to be composed of many threads — threads to memories, to others, to objects — and these relations persist in states in which our own subjectivity falls away, like sleep or death. (Taliyan 620)

A turning point of the film is the death of Lui. After feeling unwell due to a fall, he is rushed to the hospital. At the hospital, Lui lies in a hospital bed and as he is connected to the ventilator, he passes away. Elle's screen shows her waiting patiently as he is dying outside his hospital room, looking confused. Lui's half of the screen fades to black, leaving only Elle's disoriented perspective.

Following her husband's death and her refusal to take medication, Elle dementia becomes more severe. This is reflected visually in the shift of perspective: where Lui's screen once was, the perspective now belongs to their son. This persists until Elle herself dies, and finally, only the son's view remains on screen as he plans the funeral, still framed in the same split- screen structure.

Comparative Analysis

While *The Father* and *Vortex* depict dementia through radically different aesthetic lenses, both films employ cognitive disorientation as a central formal strategy- constructing not only narratives about dementia but experiences of dementia for the viewer. These works simulate cognitive decline through distinct cinematic methods: *The Father* leans toward psychological precision and theatrical polish, while *Vortex* immerses viewers in raw, observational realism. Yet despite their differences in tone and form, both demand heightened cognitive engagement from the audience, forcing them to reconstruct fractured storylines and inhabit unstable subjectivities.

In *The Father*, disorientation is crafted through narrative ambiguity and spatial instability. The polished mise- en- scène and recursive structure position the viewer within Anthony's subjectivity. The audience must actively reconcile conflicting information- shifting character identities, altered decor, and repetitive scenes- mirroring the labor of someone struggling with memory and perception.

Vortex, in contrast, does not organize disorientation around a single perspective but splits it- both literally and narratively. Gaspar Noe's use of split screen creates simultaneous yet non-synchronous experiences, emphasizing the couple's cognitive and emotional drift. Unlike *The Father's* self-contained psychological maze, *Vortex's* long takes, low lighting, and observational camera style draw from phenomenological realism. Viewers are not positioned as cognitive surrogates for the characters but are made to *feel* the unbearable slowness of time, the fragmentation of attention, and the erosion of intimate bonds. The discomfort is embodied rather than decoded. As Stadler argues, realism affects the sensorial aspects of viewing, as we believe it is more closely tied to true human experience.

The naturalistic style of social realism is presumed to have the capacity to reveal a more authentic view of the human condition and to reflect more deeply on social and political problems. (Stadler 441)

The films also differ in their relationship to temporality. *The Father* loops and collapses time through unreliable repetition, compelling the viewer to constantly revise their mental timeline. *Vortex*, on the other hand, unfolds in real time or near real time, trapping the audience in an unbearably linear progression of decline. In both, however, time ceases to function as a narrative anchor; instead, it becomes another site of confusion and emotional fatigue.

Both films place the viewer in a position of cognitive instability. Rather than offering an explanatory framework for dementia, they enact its epistemological uncertainty. They destabilize perception, identity, and environment to simulate a condition marked by fragmentation. As such, they challenge the viewer to reconstruct meaning from disorder. Whether through the artifice of *The Father* or the realism of *Vortex*, these films turn cognitive disruption into a conduit for empathy, forcing viewers to resonate with the experiences of dementia.

Conclusion

The Father and *Vortex* employ radically different formal strategies, yet both succeed in rendering the disorienting and affective reality of dementia as a perceptual experience for the viewer. Through recursive scenes, shifting spatial cues, and altered character appearances, *The Father* externalizes cognitive deterioration, placing the spectator in a state of narrative and spatial uncertainty. *Vortex*, by contrast, uses real-time split-screen imagery and minimal editing to emphasize fragmentation, simultaneity, and isolation, immersing viewers in the sensory fatigue and alienation of decaying minds. Drawing from cognitive film theory and neuroscience, this paper has shown how each film challenges traditional narrative clarity and viewer comprehension to simulate the perceptual and emotional disintegration associated with dementia. Rather than merely illustrating cognitive decline, *The Father* and *Vortex* invite the audience to feel it, highlighting cinema's unique capacity to embody invisible mental states and foster empathy. Ultimately, these films demonstrate how the medium can bridge the gap between subjective affliction and collective understanding, making the interior world of dementia perceptible, affective, and deeply human.

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